

Assignment Sheet

(Session: 2023-24)

Assignment No	string	Subject	string
Branch	string	Semester	string
Unit	Multiple Units	Faculty	string
Date Given	string	Submission Date	string

Q.No	Questions	CO	Marks
Q1	Using TensorFlow's Data API, write a code snippet to create a dataset from a CSV file containing time series data. Explain how to preprocess the data to normalize the values and convert the time series into fixed-length sequences	CO1	2
Q2	Consider a time series dataset that records the daily closing prices of a stock. Describe how you would create a windowed dataset that slides over this time series with a window size of 7 days and a shift of 1 day, ensuring that there are no missing windows	CO2	2
Q3	Given a dataset with missing time series data points, outline a method to impute these missing values using forward fill and backward fill strategies before batching the data for model training	CO1	2
Q4	Explain the concept of caching the dataset in TensorFlow and provide code to cache the data from a CSV file after the initial preprocessing steps	CO2	2

Department of Artificial Intelligence

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